



SIGMA

Delivering next generation life saving therapies

Leadership Team



Zachary Rapp

CEO & Co-Founder, Board of Directors



Chris Pihl

Co-Founder, Board of Directors



Matthew Scholz

Co-Founder, Board of Directors



Investors



Healthspan
Capital



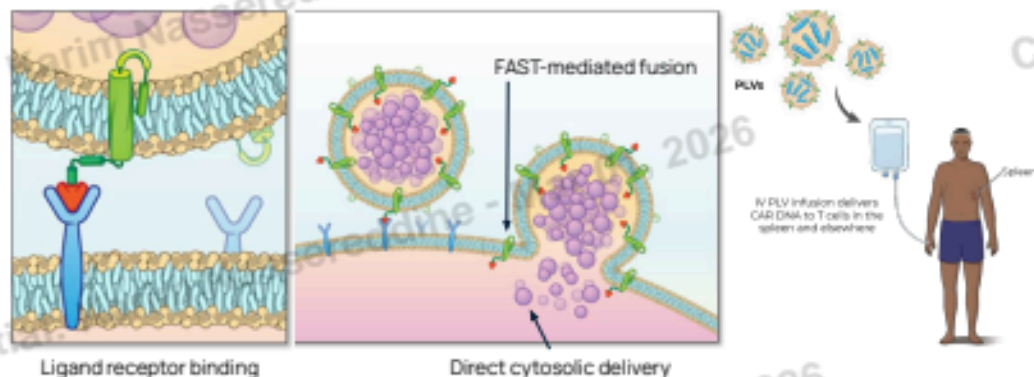
Intracellular delivery is the
bottleneck to life saving
cell and gene therapies

Problems with Current Methods

- ⚠ **Viral Vectors:** Toxicity & patient deaths remain a hurdle. Biogen, Vertex, Takeda, Pfizer wound down programs in 2025
- ⚠ **Lipid Nanoparticles:** Limited to liver targeting and vaccines. Minimal endosomal escape efficiency
- ⚠ **Electroporation:** Kills cells and doesn't scale

Best-in-class gene delivery platforms to develop novel cell therapies

Proteolipid vehicles (PLVs): In situ gene delivery for CAR-T



AI-designed FAST protein fusion delivers cargo directly into cells

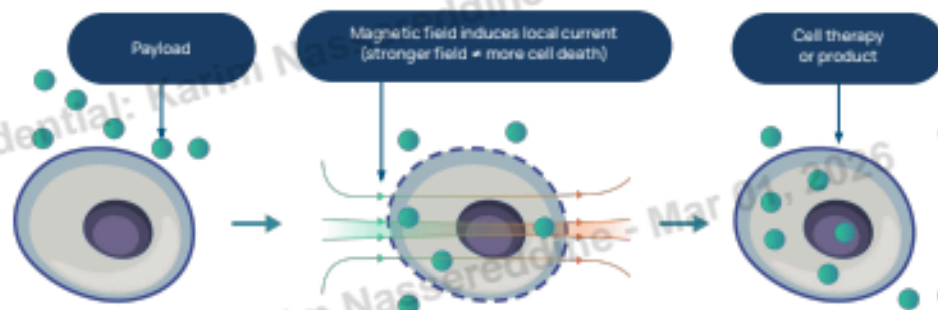
Extrahepatic delivery vs. liver (AAV/LNP): lung, CNS, muscle, eye

Non-immunogenic: enables repeat dosing

Delivers large DNA & RNA payloads

Lead: In situ dual & single CAR-T for pan solid tumors

Magnetoporation: ex vivo gene delivery for cell therapy/products



Buffer-free poration via pulsed magnetic fields

Scalable, high viability and gene delivery of DNA, RNA, and proteins

Faster manufacturing and lower COGS = broader access

Growing data moat across human & stem cell applications

Lead: ex vivo cell therapy and stem cell (hTSC) producer cell line

Our Model: Partner and in-license clinically validated therapies from leading institutions to rapidly advance life saving programs from concept to clinic

Pipeline

Product

Partners

Oncology (PLVs)

Dual CAR-T (IL13Ra2/HER2)

Single CAR-T Expansion Indication



City of
Hope

OncoSenX™

Oncology (Magnetoporation)

Platform Enablement
CAR-T Therapies



Seattle Children's
HOSPITAL · RESEARCH · FOUNDATION



Andy Hill CARE Fund
Washington's Cancer Research Endowment



ScaleReady.

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Human Trophoblast Stem Cells
Producer Cell Line



Accelerated
Biosciences

Fusogenix Proteo-Lipid Vehicle (PLV) Platform

Next generation genetic medicine delivery platform for diverse payloads

Sept 19th issue of Cell

Cell

CellPress
OPEN ACCESS

Article

Safe and effective *in vivo* delivery of DNA and RNA using proteolipid vehicles

- Fusion-mediated delivery by FAST proteins bypasses the endosomal pathway
- Cargo is delivered intracellularly compared to < 1% for endosomal escape
- Fusion entry significantly reduces immunogenicity of RNA and DNA cargos

FAST protein catalyzes rapid lipid mixing between the PLV and target cell membrane

FAST
protein

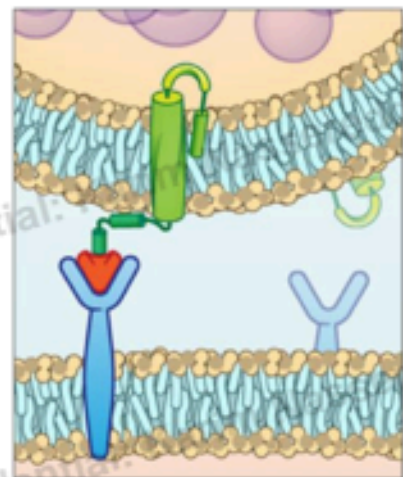
PLV cargo (plasmid DNA, mRNA, siRNA, gene editing tools, or a combination of these) is deposited directly into the cytoplasm – bypassing the endocytotic pathway

Host cell

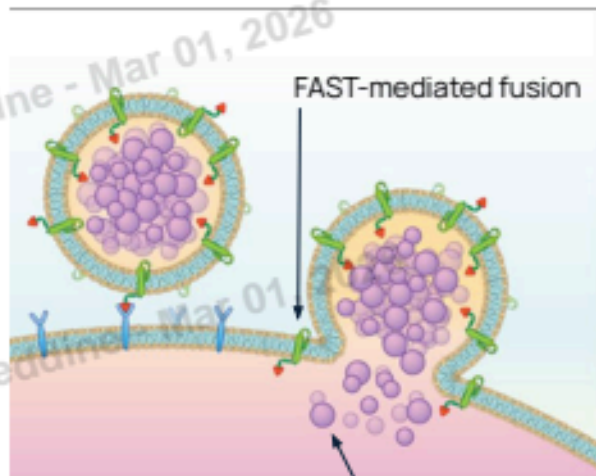
OncoSenX™

Fusogenix AIM: Precise Cell & Tissue Targeting

Fusogenix Advanced Interactive Modules (AIM) enable selective, ligand-directed PLV fusion



Ligand receptor binding



Direct cytosolic delivery

- **Targeted fusion:** AIM engineered peptide ligands or scFvs on FAST proteins to recognize specific cell surface receptors
- **Modular specificity:** Large FAST ligand library enables precise cell and tissue level targeting
- **Programmable delivery:** PLVs fuse only with receptor positive cells, enabling selective genetic payload delivery
- **Advanced capabilities:** AIM ligands unlock hard targets, including BBB transit and immune privileged tissues

In Vivo Gene Delivery is the Bottleneck in Genetic Medicine

Feature	PLV	LNP (Lipid Nanoparticle)	AAV (Viral Vector)
Delivery of DNA	✓	✗	✓
Delivery of RNA	✓	✓	✗
> 4 kb cargo	✓	✓	✗
Extrahepatic delivery	✓	✗	✗
Safety & Immunogenicity	✓	✗	✗
Repeat dosing	✓	✓	✗
Low COGS	✓	✓	✗

The Delivery Gap

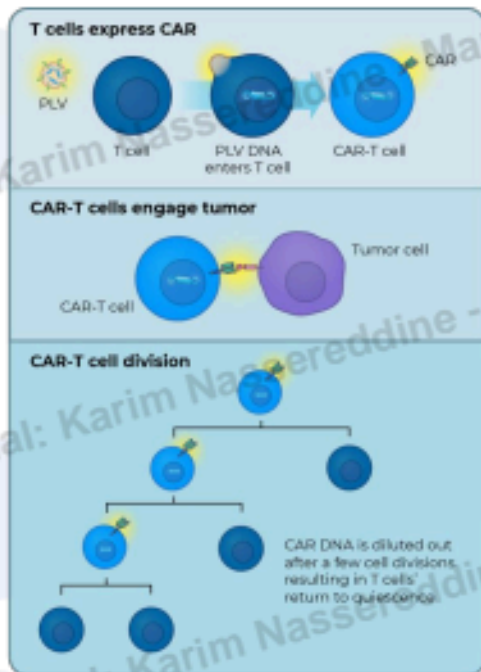
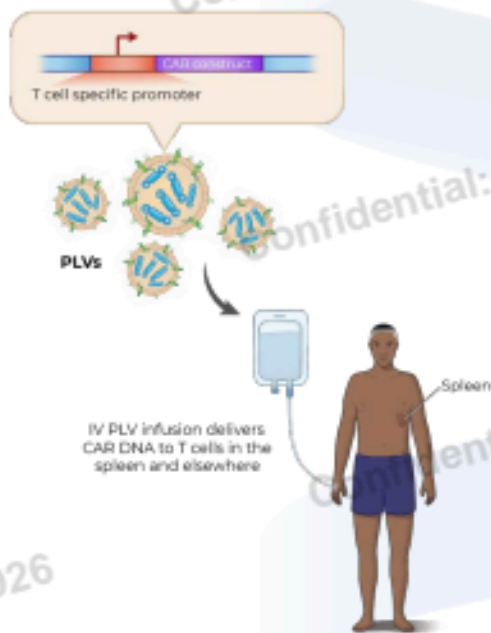
- 7 approved AAV therapies but toxicity and patient deaths remain a hurdle
- 5 approved LNP siRNA therapies all limited to liver or vaccines
- PLVs combine large cargo capacity, systemic extrahepatic delivery, non-immunogenic re-dosing, and scalable manufacturing

Re-dosable in situ CAR-T Therapies

Scalable, controllable CAR-T delivered directly in vivo

PLV-delivered CAR DNA delivers:

- **In vivo CAR-T programming:** IV dosing delivers CAR DNA directly to patient T cells
- **Targeted activation & control:** promoter/repressor logic confines CAR activity to intended cells
- **Programmable payloads:** large DNA capacity adds modules (e.g., cytokines + chemokine receptors) to drive trafficking to distal tumors
- **Self-limiting & re-dosable:** non-integrating DNA dilutes with division, limiting long term toxicity
- **Non-immunogenic:** PLVs allow re-dosing as needed



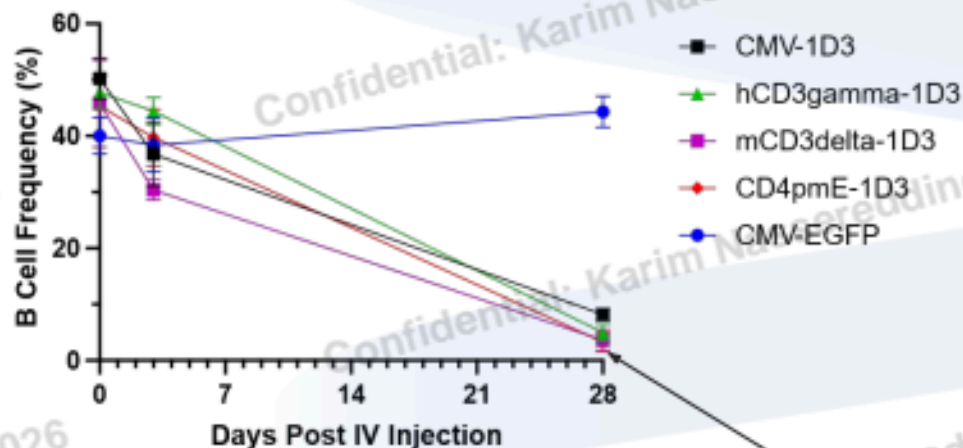
Preliminary Data

In vivo CAR Delivery Drives Functional CAR-T Activity

PLVs effectively deliver CAR-T pDNA constructs to T cells *in vivo*

- Initial GFP reporter studies displayed robust expression in immune cells of the lung and spleen, including CD4⁺ and CD8⁺ T-cell populations
 - Lung:** 7% of CD8⁺ and 33% of CD4⁺ T cells
 - Spleen:** 5% of CD8⁺ and 12% of CD4⁺ T cells
- Developed four different 1D3 CAR-T constructs targeting mouse CD19 under distinct promoters
- Post-IV administration, all CD19 CAR-T candidates produced significant depletion of B cells in spleen and PBMCs

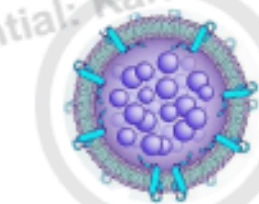
B Cell Frequency PBMCs Over Time



PBMC B cells fall to 0-10% by day 28

PLVs Enable Tissue-Specific Targeted Delivery

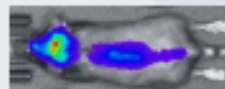
Proprietary PLV formulations can be optimized to target any tissue



PLVs are optimized for

- Lipids
- Z-average
- Surface charge
- Compressibility
- Surface decoration

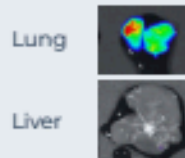
CNS Formulation
(intrathecal)



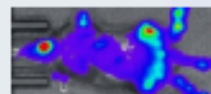
Eye Formulation
(intravitreal)



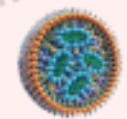
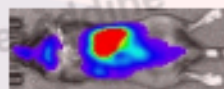
Lung Formulation
(intravenous)



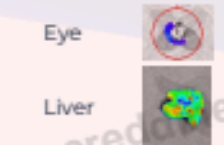
Muscle Formulation
(intravenous)



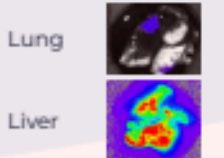
(intrathecal)



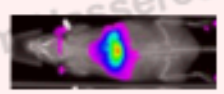
(intravitreal)



(intravenous)



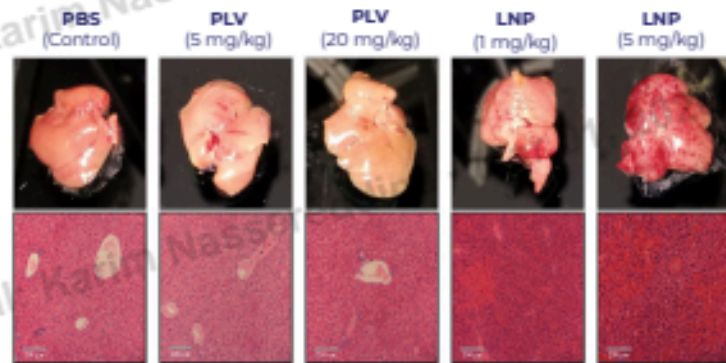
(intravenous)



PLVs Demonstrate Safe, High-Dose, Extrahepatic Gene Delivery

Excellent Systemic Tolerability

Systemic pDNA delivery: PLV vs. LNP

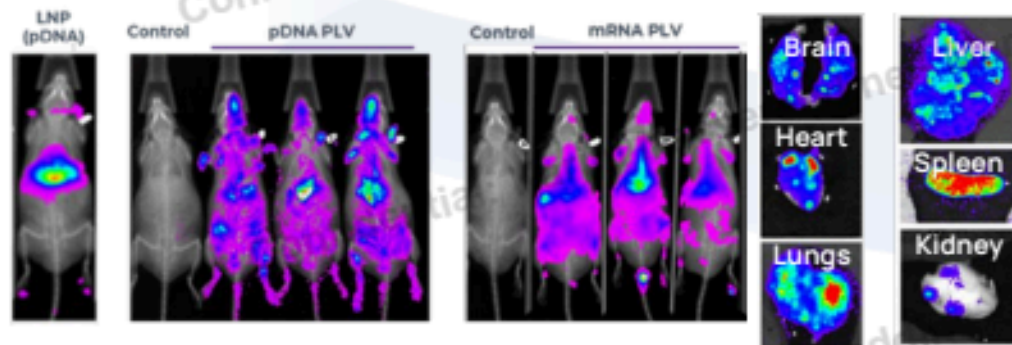


PLV doses well tolerated

LNP-treated animals died

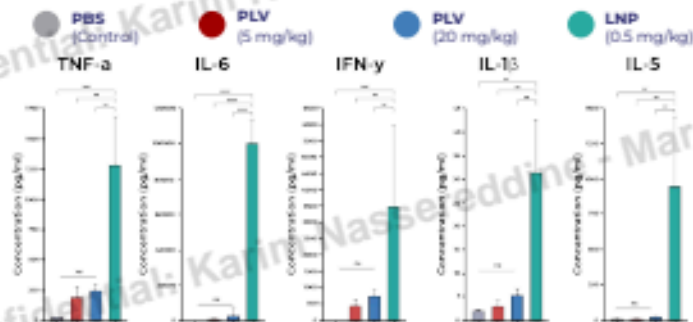
Biodistribution Beyond the Liver

Systemic pDNA and mRNA delivery: PLV vs. LNP



40× higher tolerated dose vs LNP (pDNA)

No significant systemic cytokine activation after IV dosing



Extrahepatic Biodistribution in NHPs (IV, 1 mg/kg pDNA-PLVs)



Clinically Validated Targets for Pan-Solid Tumor CAR-T

TARGET PROGRAMS

Lead program: Dual Target CAR-T

IL13Ra2 + HER2

- Reduces tumor heterogeneity and antigen escape
- Clinically de-risked antigens with broad solid-tumor relevance
- Single target follow-ons accelerate expansion in high expression tumors

City of Hope origin & IP

- **Exclusive license:** IL13Ra2 & HER2 CAR-Ts from City of Hope
- Collaborating investigators:

Scientific Lead



Christine Brown PhD
Deputy Director,
T Cell Therapeutics
Research Laboratory
City of Hope

Clinical Lead



Lisa Feldman MD PhD
Assistant Professor,
Neurosurgery &
Immunotherapy
City of Hope

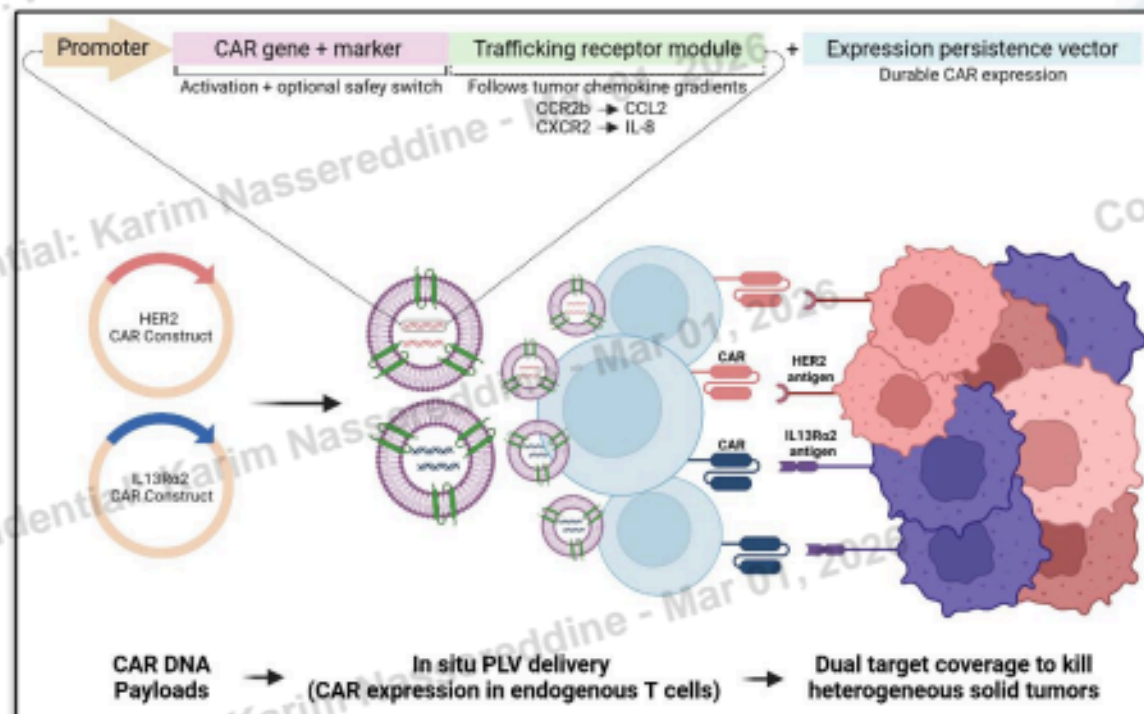
INDICATION LANDSCAPE

Targets	Oncology Indications	US Incidence/yr	IL13Ra2	HER2
Both	Glioblastoma (GBM)	11k	75%	80%
Both	Gastric	30k	50%	10-20%
Both	Ovarian	21k	32%	35%
IL13Ra2	Kidney (RCC)	81k	53%	
IL13Ra2	Pancreatic (PDAC)	67k	70%	
IL13Ra2	Prostate	314k	47%	
IL13Ra2	Colorectal	154k	66%	
IL13Ra2	Head & neck	60k	33%	
IL13Ra2	Melanoma	105k	32%	
HER2	Uterine serous carcinoma (USC subtype)	69k		35%
HER2	Medulloblastoma	1/100k		40%
HER2	Salivary duct carcinoma (SDC subset)	3k		83%
HER2	Lung (NSCLC)	225k		2-35%
HER2	Bladder (urothelial)	85k		9-61%

Next Generation In Situ PLV Dual CAR-Ts for Solid Tumors

Built for efficacy, coverage, and controllable expression

MECHANISM OF ACTION



CAR-T DESIGN ADVANTAGES

Re-dosable safety control

PLV delivery of non-integrating CAR DNA for transient expression and ability to re-dose, enabling controllable activity and a self-limiting safety profile

Trafficking to tumors

Enable CAR-T cells with chemokine-guided trafficking modules to follow tumor chemokine gradients and infiltrate distal tumor sites

Dual-target coverage

Dual targeting of heterogeneous tumors reduces relapse driven by antigen escape

Superior persistence when needed

When longer activity is required, we can add an expression persistence module to extend transgene expression without changing the delivery approach

Dual CAR-T Beachhead: \$60B annual TAM

Global TAM = annual incidence × \$50k/dose (Dual CAR-T pDNA + PLV encapsulation)

GBM

\$8.3B

Global TAM (annual)

168K

Global Incidence / yr

626K

Global Prevalence

Gastric

\$35B

Global TAM (annual)

968K

Global Incidence / yr

1.63M

Global Prevalence

Ovarian

\$16.2B

Global TAM (annual)

325K

Global Incidence / yr

930K

Global Prevalence

Corporate Relationship & Licenses

Partnering with leading translational research institutions, we in-license clinical validated breakthrough assets to rapidly advance best-in-class therapies to patients



(Platform Licensor)

- Late stage preclinical company exclusively developing Proteo-Lipid Vehicles (PLVs) in oncology
- Payload-agnostic delivery platform (CAR-T, suicide gene)
- Grants Sigma Genetics exclusive for PLVs in oncology (CAR-T antigen and indication specific)



SIGMA

(Pre-IND & Manufacturer)

- In-license PLVs for in vivo CAR-T from OncoSenX and CAR-T candidates from City of Hope
- Engineer next-gen CAR-T plasmids and PLV formulations to perform in vitro/in vivo preclinical validation and GLP tox, IND filings
- Manufacture PLVs for CAR-T programs (research and GMP grade)



City of Hope

(CAR-T Licensor)

- Academic partners with deep R&D track record in CAR-T therapy, provides translational know-how
- Exclusively license CAR-T therapies for solid tumors and hematologic malignancies to Sigma Genetics
- Provides support in pre-IND R&D, IND filing, and clinical trials

PLV Platform License & Formulation



CAR Vector / PLV Design & Preclinical Development



Target Discovery & Clinical Translation

Pipeline

Product

Partners

Oncology (PLVs)

Dual CAR-T (IL13Ra2/HER2)

Single CAR-T Expansion Indication



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OncoSenX™

Oncology (Magnetoporation)

Platform Enablement
CAR-T Therapies



Seattle Children's
HOSPITAL · RESEARCH · FOUNDATION



Andy Hill CARE Fund
Washington's Cancer Research Endowment



ScaleReady.

Cell Line Development (Magnetoporation)

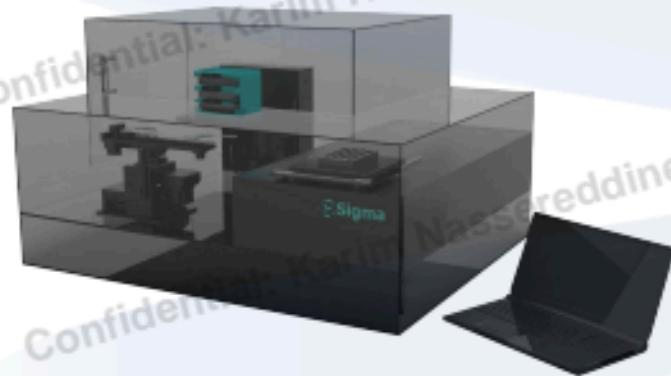
Human Trophoblast Stem Cells
Producer Cell Line



Accelerated
Biosciences

Next Generation, Scalable Breakthrough: Magnetoporator™

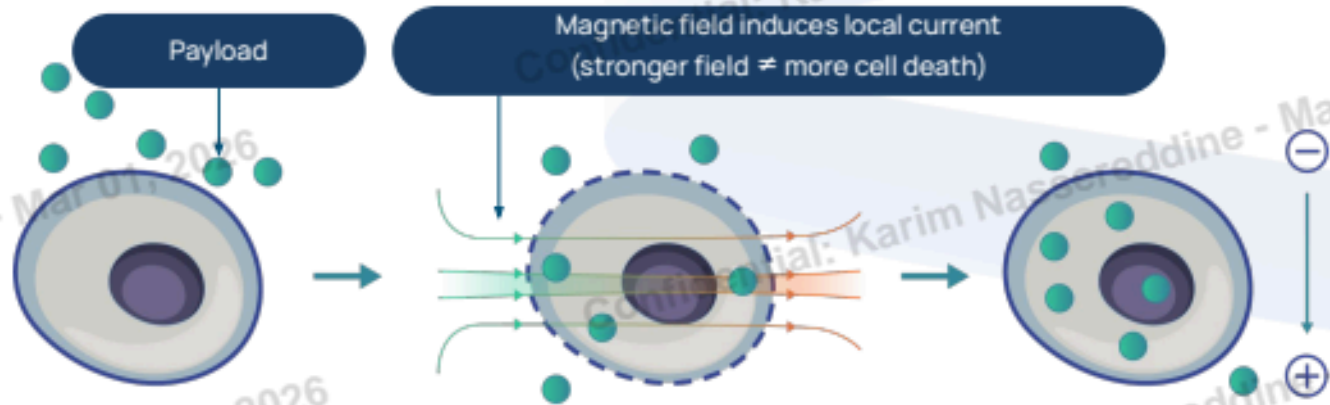
- ✓ **Patented platform:** Electrode-free poration using pulsed magnetic fields – protected IP in a rare whitespace of intracellular delivery
- ✓ **Uniform exposure:** Eliminates arcing and uneven field distribution, enabling consistent delivery without use of buffers or beads
- ✓ **Improved cell quality:** Minimizes electrical stress and damage, delivering superior cell survival and proliferation
- ✓ **Controlled field generation:** Precision waveform circuits (pulse generators, oscillators) = reproducible delivery with diverse payloads
- ✓ **Enhanced penetration:** Optimized coil configurations facilitate deep, uniform field distribution – ideal for clinical scale
- ✓ **Broad compatibility:** Supports delivery of DNA, RNA, proteins, and cryoprotectants for drug discovery and therapeutic development



Magnetoporation vs. Electroporation

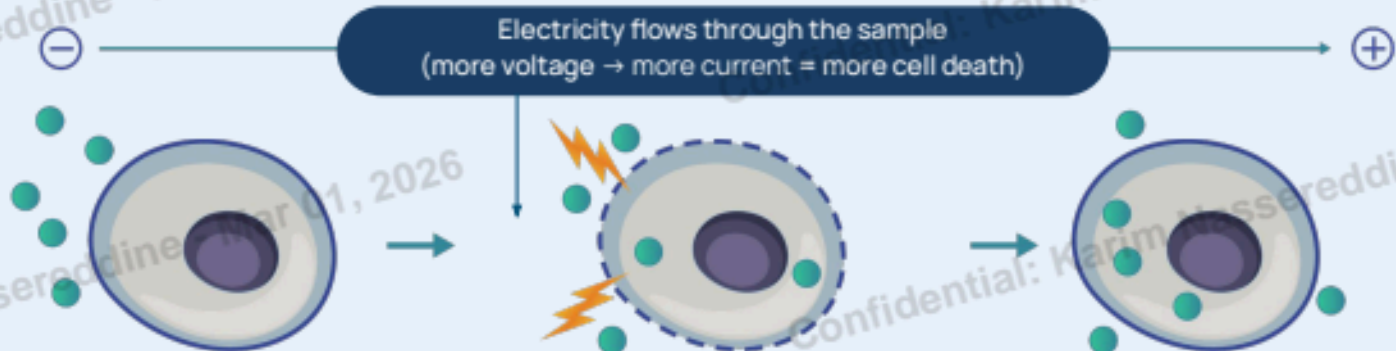
Magnetoporation

nanosecond pulsed electric field (nsPEF)

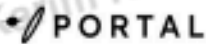


Electroporation

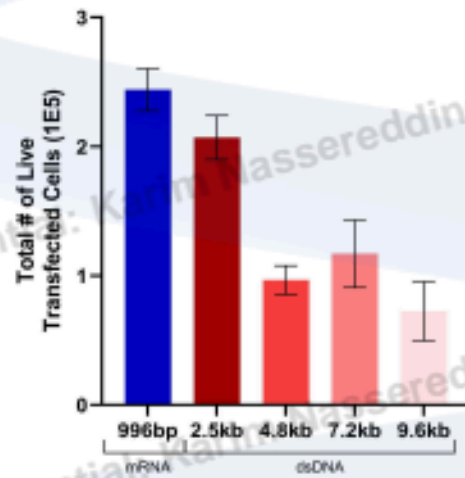
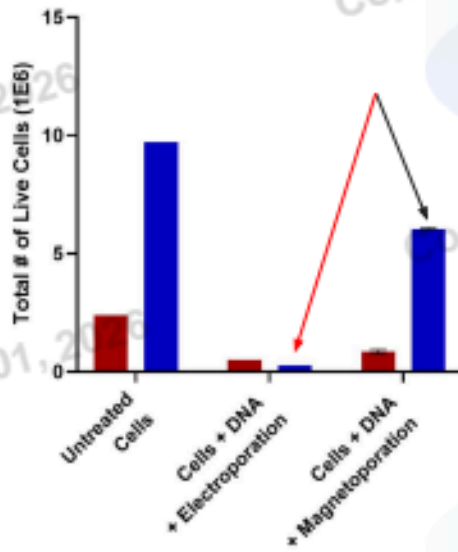
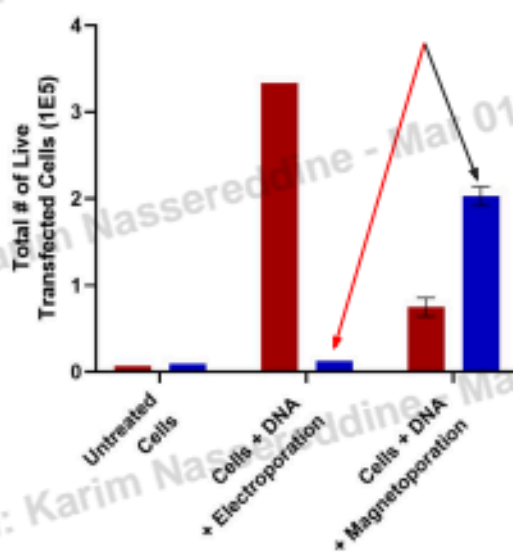
microsecond pulsed electric field (μ sPEF)



A Closer Look at Ex Vivo Gene Delivery Methods

	Magnetoporation	Viral Vectors	Electroporation	Mechanoporation
Cell Survival	✓	✓	✗	✓
Payload flexibility (size & type)	✓	✗	✗	?
Non-immunogenic	✓	✗	✓	✓
Scalability / COGS	✓	✗	✗	?
Companies	 SIGMA	 Bristol-Myers Squibb  GILEAD  bluebird bio  Oxford Biomedica  NOVARTIS	 Milenyi Biotec  MaxCyte  Lonza  BIO-RAD  inovio  ONCOSEC  ThermoFisher Scientific  BTX	 indee labs  KYTO PEN  CellFE  PORTAL

Superior Expansion & Transfection in Primary CD4 T Cells



- Magnetoporation delivers 16× more transfected and 22× more live cells vs. electroporation at 96h
- Efficient delivery of mRNA and dsDNA up to 9.6 kb
- Consistent performance across multiple donors

Magnetoporation for CAR-T Cell Therapy Manufacturing

Magnetoporation (MP)

\$2.5-5K per patient

Enable more efficient gene delivery:

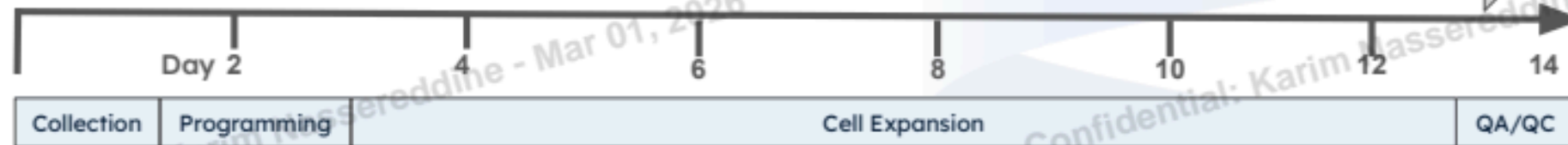
- Less expensive than LVV & EP (time & materials)
- No specialized reagents or buffers required
- Faster cell proliferation = decrease cell expansion period
- Target transfection + cell expansion in 72 hrs

Lentivirus (LVV)

+/- \$30K per patient

LVV & EP both result in longer cell expansion periods. LVV manufacturing is expensive, lengthy (≥12 month delivery) with safety concerns. EP requires specialized reagents and buffers.

Electroporation (EP)



Business Model

Customers
→ Academic labs → Biotech/Pharma → Research hospitals → CROs/CDMOs

Collaborators
→ \$25k per collaboration → Research use only (RUO) device → \$50k-\$30k per device & cuvette revenue → 90% device margin

Clinical Trials
→ \$100k per cGMP device → \$2.5k-\$5k patient dose fee (Phase 1 to FDA approved drug) → Full tech support

FDA Approved
→ DMF Fee to FDA → \$7500 per patient dose → % of therapy gross annual sales

Pre-clinical Development and Clinical Trials

FDA approved Drug

Market Opportunity: T cell Therapies

200

Research device sales
by 2027

10

Clinical device sales for
Phase 1 clinical trials

\$9M

Revenue per CAR-T up
to Phase 3 clinical trials

16M

US/EU Cancer and
Autoimmune patients
by 2030

\$7500

Revenue per
patient

\$120B

TAM by 2030

Pipeline

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Biosciences

Human Trophoblast Stem Cells (hTSCs)

Ethically derived, stem cells for the future of regenerative therapeutics and biomanufacturing

Origin

- Derived from trophoblast (placenta-forming outer layer of the blastocyst)
- Ethically sourced, consented non-embryonic tissue (no embryo or fetal destruction)

Key Biological Features

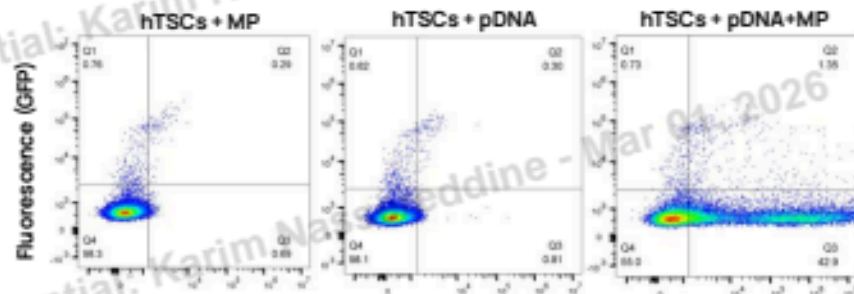
- Self-renewing = supports long-term culture
- Multilineage differentiation capacity
- Genetically stable and non-tumorigenic
- Immune-privileged (low rejection risk)

Opportunity

- Develop superior producer cell line for drug and cell product manufacturing

Proof of Concept

- Magnetoporation of hTSCs with pDNA-eGFP
- Achieved up to 40% transfection efficiency with minimal cell loss



Blastocyst



Trophoblast layer



Scale cell culture

hTSC Platform to Replace Legacy Fetal Cell Lines

Problem: Legacy Cell Lines

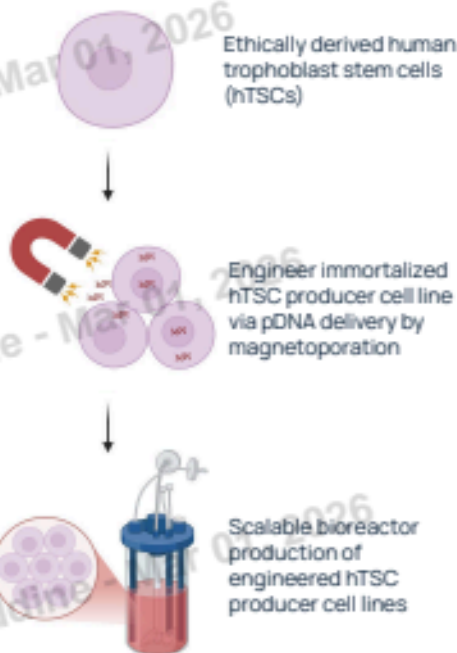


HEK293

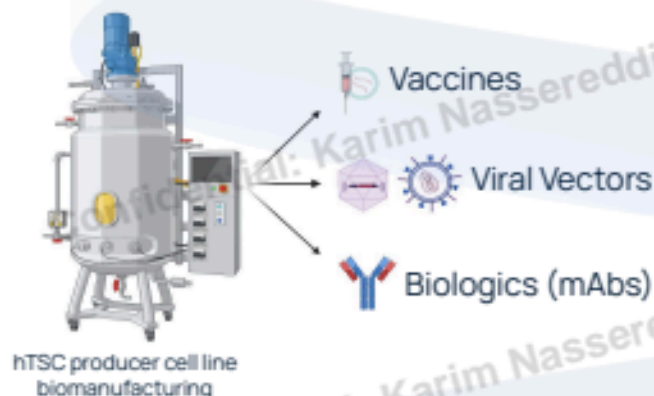
PER.C6

- Legacy fetal-derived platforms
- Adenoviral oncogene integration
- Genomic instability risk
- Ethical and regulatory constraints

Solution: hTSC Platform



Market Opportunity: hTSC Producer Line



Sigma's position: hTSCs offer an ethical, genomically stable, immune privileged replacement for HEK293 and PER.C6 for next generation biomanufacturing.

Latest Transaction: [JSJ acquired Crucell \(PER.C6 platform\) for \\$2.4B](#) to secure a proprietary vaccine and biologics producer cell line.

Business Model + TAM

Licensing led platform for next generation hTSC producer cell lines

Customer: CDMOs & Pharma

Platform license

- Upfront access to hTSC producer cell line(s) and field of use commercial rights



Custom producer line engineering

- Stable integration, clone selection, characterization, productivity optimization



GMP cell bank and tech transfer

- Master cell bank (MSB / Working cell bank (WCB), package transfer to CDMO/Pharma



Commercial royalties

- Per product royalties / per batch fees

Revenue Model

Upfront + annual

Milestones

Per program

Recurring

\$11B+ TAM

2024 global spend where we monetize

\$5.7B

Cell line development

Producer lines for biologics (ex. mAb)

\$5.2B

Viral vector manufacturing

AAV, LVV, Adv

End
Market
Pull

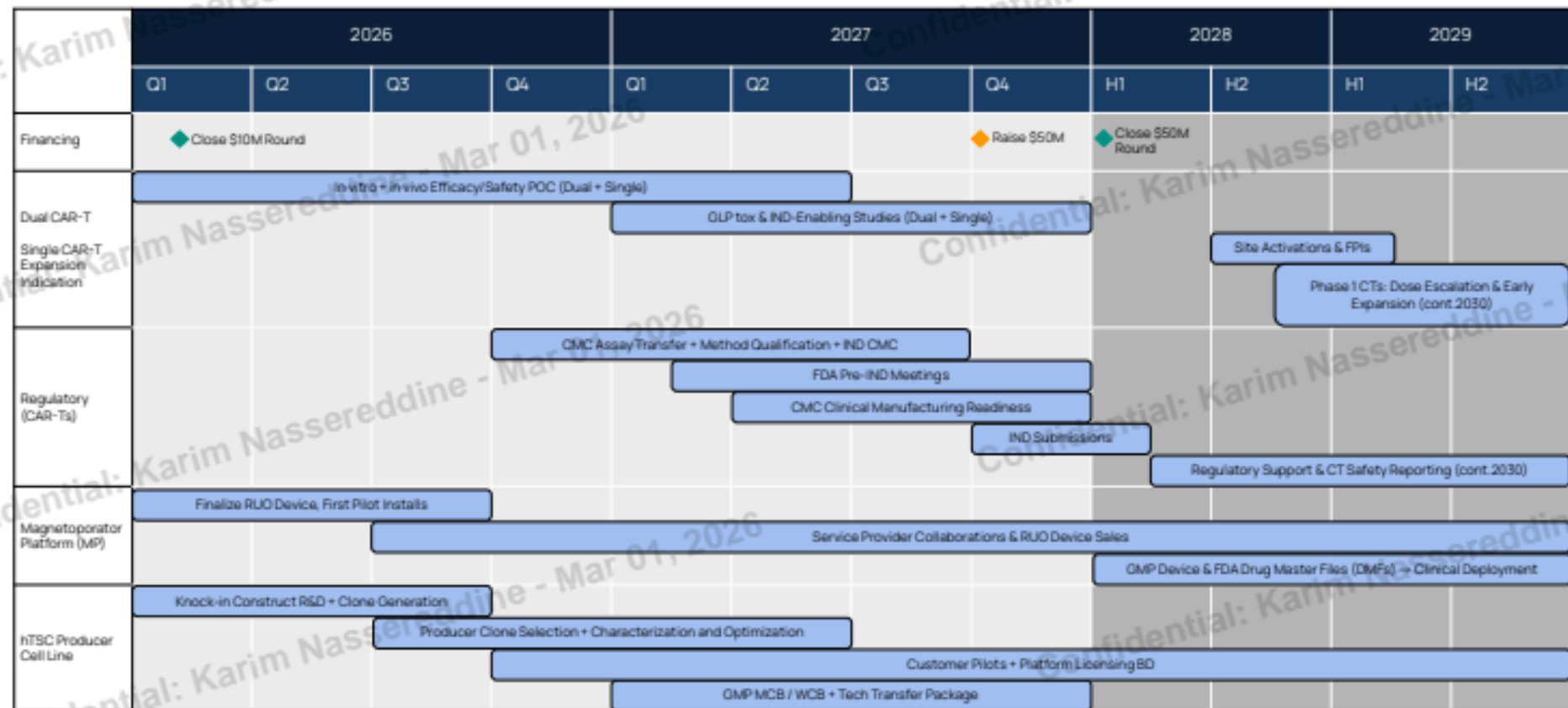
\$77B Vaccines (2023)

\$276B mAbs (2024)

Large installed base of HEK293 and PER.C6

\$10M Raise: Milestones & Timeline

Use of proceeds: 2x CAR-T IND-enabling → IND submissions. CMC/analytics. RUO → GMP Magnetoprotor (DMFs → clinical deployment). hTSC producer line (GMP MCB/WCB + customer pilots/licensing).



*\$2M Seed+ funds pre-\$10M round deliverables through \$10M close

Contact Us

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